Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester: II* | *Time: 03 Hours*
Code: 100203 | *Subject: Chemistry* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following: **[2 x 7 = 14]**

(a) What is an orbital?
(b) What is selection rule?
(c) What is the condition for a reaction to be in equilibrium?
(d) Transition metal ions like Cu^{+} and Ag^{+} are colourless. Why?
(e) State de Broglie's principle.
(f) The presence of CO_2 in boiler feed water should be avoided. Why?
(g) What is critical temperature of a gas?
(h) What is the direction of a reaction when $\Delta G = 0$?
(i) Arrange the covalent bonds $C-F, C-Br, C-Cl$ and $C-I$ in increasing order of strength.
(j) Which of NH_3, PH_3, AsH_3 and SbH_3 possesses largest bond angle? Give reason.

Q.2 (a) Which of the following two molecules has a higher bond length? **[6]**

(i) O_2

(ii) O_2^+

(iii) O_2^-

Explain using molecular-orbital theory.

(b) Draw the MO energy-level diagram for NO molecule. Using this diagram, calculate and explain bond orders and magnetic behaviours of (i) NO, (ii) NO^+ (iii) NO^- **[8]**

Q.3 (a) The internuclear distance of NaCl is $2.36 \times 10^{-10} m$. Calculate the reduced mass and moment of inertia of NaCl (atomic mass of Cl = 35×10^{-3} kg mol^{-1} and Na= 23×10^{-3} kg mol^{-1}). **[4]**

(b) Calculate the force constant for CO, if it absorbs at 2.143×10^5 m^{-1} (atomic mass of C=12 $\times 10^{-3}$ kg mol^{-1} and O=16$\times 10^{-3}$ kg mol^{-1}). **[4]**

(c) Define chromophore and auxochrome with examples. **[4]**

(d) Which type of electronic transition(s) is/are observed in UV-visible spectrum of propanone in the range 200 nm to 900 nm? Justify your answer. **[2]**

Q.4 (a) 2 mole of NH_3 at 300 K occupy a volume of $5 \times 10^{-3} m^3$. Calculate the pressure using van der Waals equation (a=0.147 N m^4 mol^{-2} and b= 0.037×10^{-3} m^3 mol^{-1}). Compare the above result with the pressure calculated using ideal gas equation. **[5]**

(b) Write short notes on the following: **[9]**

(i) Magnetic resonance imaging
(ii) Fingerprint region in infrared spectroscopy
(iii) Different types of electronic excitations

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Q.5 (a) Consider the following half-cell reactions: [7]

$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e \rightarrow PbSO_4(s) + 2H_2O$, $E^\circ = 1.70$ V

$PbSO_4(s) + 2e \rightarrow Pb(s) + SO_4^{2-}(aq)$, $E^\circ = -0.31$ V

Write the cell (in proper cell notation) and the cell reaction. Calculate the value of E° for the cell and the EMF generated if $[H^+] = 0.1\ M$ and $[SO_4^{2-}] = 2\ M$.

(b) A water sample has the following constituents per litre: [7]
$CaCO_3$ = 81 mg, $Mg(HCO_3)_2$ = 75 mg. $CaSO_4$ = 136 mg, $MgSO_4$ =120 mg. NaCl = 4.7 mg
Calculate the quantity of temporary and permanent hardness in the water sample. Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 1.5 million litres of the above water sample.

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Q.6 (a) Write notes on the following : [7]
(i) Optical isomerism of lactic acid
(ii) Optical isomerism of tartaric acid

(b) Differentiate between the following: [7]
(i) Enantiomers and Diastereomers
(ii) Racemic mixture and Meso-compounds

Q.7 (a) Explain the following : [7]
(i) Trichloroacetic acid is stronger acid than acetic acid.
(ii) The amino group in aniline is *o*- and *p*- directing but nitro group is *m*-directing.

(b) Arrange the following carbocations in order of increasing stability with suitable reasons: [7]
$C_6H_5CH_2^+$, $(CH_3)_3C^+$, $(C_6H_5)_2CH^+$, $CH_3CH_2^+$

Q.8 (a) Write short notes on the following: [7]
(i) Wave-particle duality
(ii) Heisenberg's uncertainty principle

(b) Discuss photoelectric effect (with mathematical equations) for explaining the particle nature of light. [7]

Q.9 Write short notes on ***any two*** of the following:- [7x2=14]
(a) Nuclear magnetic resonance and its application
(b) Markownikoff 's rule and Kharasch's rule
(c) Steric effects and Diels-Alder reaction

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